Brightloop Subscription Management API

Postman test case document and execution results

401 test cases in **17** modules, executed with newman against the FastAPI backend (SQLite demo database, fresh seed). Result: **401 passed, 0 failed**; 1,154 assertions, 0 failures. Average response time 12 ms, maximum 156 ms (target: under 2 s). PASS* = case needs an e-mail token printed in the server console; executed in a second pass with real tokens.

Module	Cases	Passed
00 Health and Root	2	2
01 Authentication	28	28
02 User Management (Admin)	22	22
03 Customers	12	12
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09 Subscriptions	47	47
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11 Billing and Invoices	62	62
12 Payments	35	35
13 Notifications	11	11
14 Reports and Dashboard	10	10
15 Security and Access Control	20	20
16 E-mail Token Flows (manual tokens)	4	4
Total	401	401

How to run

1. Start the backend (see README) and seed demo data: `python -m app.db.seed; python scripts/seed_demo.py`.
2. Import `Brightloop.postman_collection.json` and `Brightloop.postman_environment.json` into Postman.
3. Run the whole collection top to bottom with the Collection Runner (order matters: later cases reuse ids created earlier).
4. CLI: `newman run Brightloop.postman_collection.json -e Brightloop.postman_environment.json`.
5. Folder 16 needs tokens from the server console (dev e-mail mode): set `verify_token` and `reset_token` in the environment, otherwise those 4 cases are skipped.

Every request asserts its status code and a response time below 2 seconds; most also assert body fields and computed amounts.

Scenario data used by the suite

Customers, staff and products are created per run (unique run id), so the suite can be re-run. Key arithmetic checked: Odoo variant 1000 + 560 = 1560 per unit; quotation totals 3681.60 / 3563.60 / 1840.80; invoice A 3313.44 after the TEN discount; payments of 1000.00 + 500.00 + 1813.44 settle it; a refund reverts it to partially paid.

Notes

Discounts are applied manually by code on draft invoices (not automatically). Migrations target MySQL; this run used SQLite tables created from the models. Performance with thousands of records was not load-tested.

00 Health and Root

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-001	Health check	GET	/api/v1/health	Anonymous (no token)	Service reports healthy.	200	200	PASS
TC-002	API root	GET	/	Anonymous (no token)	Root returns app name, version and docs link.	200	200	PASS

01 Authentication

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-003	Register customer 1 (valid)	POST	/api/v1/auth/register	Anonymous (no token)	Account created with role CUSTOMER, e-mail not yet verified, no password data in the response. Body: {"full_name": "Postman Customer One", "email": "pm.{{run_id}}.c1@brightloop.demo", "password": "{{std_password}}"}	201	201	PASS
TC-004	Register customer 2 (valid)	POST	/api/v1/auth/register	Anonymous (no token)	Second customer account created (used later for ownership/isolation tests). Body: {"full_name": "Postman Customer Two", "email": "pm.{{run_id}}.c2@brightloop.demo", "password": "{{std_password}}"}	201	201	PASS
TC-005	Register - duplicate email rejected	POST	/api/v1/auth/register	Anonymous (no token)	Each email can belong to only one account. Body: {"full_name": "Dup", "email": "pm.{{run_id}}.c1@brightloop.demo", "password": "{{std_password}}"}	409	409	PASS
TC-006	Register - duplicate email, different letter case rejected	POST	/api/v1/auth/register	Anonymous (no token)	E-mail uniqueness is case-insensitive. Body: {"full_name": "Dup", "email": "PM.{{run_id}}.C1@BRIGHTLOOP.DEMO", "password": "{{std_password}}"}	409	409	PASS
TC-007	Register - password of exactly 8 characters rejected	POST	/api/v1/auth/register	Anonymous (no token)	Password must be MORE than eight characters (8 is too short). Body: {"full_name": "Weak", "email": "pm.{{run_id}}.w1@brightloop.demo", "password": "Abcde#12"}	422	422	PASS
TC-008	Register - password without uppercase rejected	POST	/api/v1/auth/register	Anonymous (no token)	Password needs an uppercase letter. Body: {"full_name": "Weak", "email": "pm.{{run_id}}.w2@brightloop.demo", "password": "strong#pass1"}	422	422	PASS
TC-009	Register - password without lowercase rejected	POST	/api/v1/auth/register	Anonymous (no token)	Password needs a lowercase letter. Body: {"full_name": "Weak", "email": "pm.{{run_id}}.w3@brightloop.demo", "password": "STRONG#PASS1"}	422	422	PASS
TC-010	Register - password without special character rejected	POST	/api/v1/auth/register	Anonymous (no token)	Password needs a special character. Body: {"full_name": "Weak", "email": "pm.{{run_id}}.w4@brightloop.demo", "password": "StrongPass12"}	422	422	PASS

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-011	Register - invalid email format rejected	POST	/api/v1/auth/register	Anonymous (no token)	Malformed e-mail address is refused. Body: {"full_name": "Bad", "email": "not-an-email", "password": "{{std_password}}"}	422	422	PASS
TC-012	Register - missing full name rejected	POST	/api/v1/auth/register	Anonymous (no token)	Required field validation. Body: {"email": "pm.{{run_id}}.w5@brightloop.demo", "password": "{{std_password}}"}	422	422	PASS
TC-013	Login customer 1 (valid)	POST	/api/v1/auth/login	Anonymous (no token)	JWT access token issued with expiry. Body: {"email": "pm.{{run_id}}.c1@brightloop.demo", "password": "{{std_password}}"}	200	200	PASS
TC-014	Login customer 2 (valid)	POST	/api/v1/auth/login	Anonymous (no token)	Second customer logs in. Body: {"email": "pm.{{run_id}}.c2@brightloop.demo", "password": "{{std_password}}"}	200	200	PASS
TC-015	Login - e-mail is case-insensitive	POST	/api/v1/auth/login	Anonymous (no token)	Login works regardless of e-mail letter case. Body: {"email": "PM.{{run_id}}.C1@BRIGHTLOOP.DEMO", "password": "{{std_password}}"}	200	200	PASS
TC-016	Login admin (seeded)	POST	/api/v1/auth/login	Anonymous (no token)	Seeded admin logs in. Body: {"email": "{{admin_email}}", "password": "{{admin_password}}"}	200	200	PASS
TC-017	Login - wrong password rejected	POST	/api/v1/auth/login	Anonymous (no token)	Generic error, no hint whether the e-mail exists. Body: {"email": "pm.{{run_id}}.c1@brightloop.demo", "password": "Wrong#Pass99"}	401	401	PASS
TC-018	Login - unknown e-mail gives same error	POST	/api/v1/auth/login	Anonymous (no token)	Same message as wrong password (no account enumeration). Body: {"email": "nobody.{{run_id}}@brightloop.demo", "password": "{{std_password}}"}	401	401	PASS
TC-019	Login - missing password rejected	POST	/api/v1/auth/login	Anonymous (no token)	Required field validation. Body: {"email": "pm.{{run_id}}.c1@brightloop.demo"}	422	422	PASS
TC-020	Current user - valid token	GET	/api/v1/auth/me	Customer 1	Returns the logged-in user.	200	200	PASS
TC-021	Current user - no token	GET	/api/v1/auth/me	Anonymous (no token)	Protected endpoint requires a token.	401	401	PASS
TC-022	Current user - malformed token	GET	/api/v1/auth/me	garbage_token	Garbage token is refused.	401	401	PASS
TC-023	Forgot password - registered e-mail	POST	/api/v1/auth/forgot-password	Anonymous (no token)	Generic confirmation. A reset link is e-mailed (printed to the server console in dev mode). Body: {"email": "pm.{{run_id}}.c1@brightloop.demo"}	200	200	PASS
TC-024	Forgot password - unknown e-mail gives identical answer	POST	/api/v1/auth/forgot-password	Anonymous (no token)	Same response for unknown address so registered e-mails cannot be discovered. Body: {"email": "ghost.{{run_id}}@brightloop.demo"}	200	200	PASS
TC-025	Forgot password - invalid e-mail format	POST	/api/v1/auth/forgot-password	Anonymous (no token)	Format validation. Body: {"email": "nope"}	422	422	PASS
TC-026	Reset password - invalid token rejected	POST	/api/v1/auth/reset-password	Anonymous (no token)	Unknown token cannot reset a password. Body: {"token": "this-token-does-not-exist-123456", "new_password": "NewStrong#Pass2"}	400	400	PASS
TC-027	Reset password - weak new password rejected	POST	/api/v1/auth/reset-password	Anonymous (no token)	New password must satisfy the same rules. Body: {"token": "this-token-does-not-exist-123456", "new_password": "weak"}	422	422	PASS

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-028	Verify e-mail - invalid token rejected	POST	/api/v1/auth/verify-email	Anonymous (no token)	Unknown verification token. Body: {"token": "this-token-does-not-exist-123456"}	400	400	PASS
TC-029	Resend verification - unverified customer	POST	/api/v1/auth/resend-verification	Customer 1	A new verification e-mail is issued.	200	200	PASS
TC-030	Resend verification - already verified user rejected	POST	/api/v1/auth/resend-verification	Admin	Admin is already verified.	400	400	PASS

02 User Management (Admin)

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-031	Admin creates staff (INTERNAL)	POST	/api/v1/users	Admin	Staff account created, e-mail pre-verified. Body: {"full_name": "Postman Staff", "email": "pm.{{run_id}}.staff@brightloop.demo", "password": "{{std_password}}", "role": "INTERNAL"}	201	201	PASS
TC-032	Create staff - duplicate e-mail rejected	POST	/api/v1/users	Admin	E-mail must be unique. Body: {"full_name": "Dup", "email": "pm.{{run_id}}.staff@brightloop.demo", "password": "{{std_password}}", "role": "INTERNAL"}	409	409	PASS
TC-033	Create staff - role CUSTOMER not allowed	POST	/api/v1/users	Admin	Only ADMIN or INTERNAL can be created here. Body: {"full_name": "X", "email": "pm.{{run_id}}.x1@brightloop.demo", "password": "{{std_password}}", "role": "CUSTOMER"}	422	422	PASS
TC-034	Create staff - unknown role rejected	POST	/api/v1/users	Admin	Role is an enumeration. Body: {"full_name": "X", "email": "pm.{{run_id}}.x2@brightloop.demo", "password": "{{std_password}}", "role": "SUPERUSER"}	422	422	PASS
TC-035	Create staff - weak password rejected	POST	/api/v1/users	Admin	Same password policy. Body: {"full_name": "X", "email": "pm.{{run_id}}.x3@brightloop.demo", "password": "weak", "role": "INTERNAL"}	422	422	PASS
TC-036	Create staff - customer forbidden	POST	/api/v1/users	Customer 1	Only the admin can add staff. Body: {"full_name": "X", "email": "pm.{{run_id}}.x4@brightloop.demo", "password": "{{std_password}}", "role": "INTERNAL"}	403	403	PASS
TC-037	Create staff - no token	POST	/api/v1/users	Anonymous (no token)	Authentication required. Body: {"full_name": "X", "email": "pm.{{run_id}}.x5@brightloop.demo", "password": "{{std_password}}", "role": "INTERNAL"}	401	401	PASS
TC-038	Login staff (valid)	POST	/api/v1/auth/login	Anonymous (no token)	New staff member can log in. Body: {"email": "pm.{{run_id}}.staff@brightloop.demo", "password": "{{std_password}}"}	200	200	PASS
TC-039	Create staff - staff forbidden	POST	/api/v1/users	Staff (INTERNAL)	Internal staff cannot add staff. Body: {"full_name": "X", "email": "pm.{{run_id}}.x6@brightloop.demo", "password": "{{std_password}}", "role": "INTERNAL"}	403	403	PASS
TC-040	List users - admin	GET	/api/v1/users?page=1&page_size=50	Admin	Paged list of users.	200	200	PASS
TC-041	List users - filter by role INTERNAL	GET	/api/v1/users?role=INTERNAL	Admin	Only INTERNAL users returned.	200	200	PASS
TC-042	List users - staff forbidden	GET	/api/v1/users	Staff (INTERNAL)	User list is admin-only.	403	403	PASS

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-043	Create temp staff for deactivation test	POST	/api/v1/users	Admin	Throw-away staff account. Body: {"full_name": "Temp Staff", "email": "pm.{{run_id}}.temp@brightloop.demo", "password": "{{std_password}}", "role": "INTERNAL"}	201	201	PASS
TC-044	Login temp staff	POST	/api/v1/auth/login	Anonymous (no token)	Token obtained before deactivation. Body: {"email": "pm.{{run_id}}.temp@brightloop.demo", "password": "{{std_password}}"}	200	200	PASS
TC-045	Deactivate temp staff	PATCH	/api/v1/users/{temp_user_id}/active	Admin	Account flagged inactive. Body: {"is_active": false}	200	200	PASS
TC-046	Deactivated user's existing token stops working	GET	/api/v1/auth/me	Temp staff	Tokens of disabled users are refused immediately.	401	401	PASS
TC-047	Deactivated user cannot log in	POST	/api/v1/auth/login	Anonymous (no token)	Clear 'account disabled' error. Body: {"email": "pm.{{run_id}}.temp@brightloop.demo", "password": "{{std_password}}"}	403	403	PASS
TC-048	Reactivate temp staff	PATCH	/api/v1/users/{temp_user_id}/active	Admin	Account enabled again. Body: {"is_active": true}	200	200	PASS
TC-049	Reactivated user can log in again	POST	/api/v1/auth/login	Anonymous (no token)	Access restored. Body: {"email": "pm.{{run_id}}.temp@brightloop.demo", "password": "{{std_password}}"}	200	200	PASS
TC-050	Admin cannot deactivate own account	PATCH	/api/v1/users/1/active	Admin	Safeguard against locking everyone out. Body: {"is_active": false}	400	400	PASS
TC-051	Deactivate unknown user	PATCH	/api/v1/users/999999/active	Admin	Unknown id. Body: {"is_active": false}	404	404	PASS
TC-052	Deactivate - staff forbidden	PATCH	/api/v1/users/{temp_user_id}/active	Staff (INTERNAL)	Admin-only action. Body: {"is_active": false}	403	403	PASS

03 Customers

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-053	Customer reads own profile	GET	/api/v1/customers/me	Customer 1	Customer record auto-created at registration.	200	200	PASS
TC-054	Customer 2 reads own profile	GET	/api/v1/customers/me	Customer 2	Second customer profile.	200	200	PASS
TC-055	Customer updates own profile	PUT	/api/v1/customers/me	Customer 1	Phone/company updated. tax_id in the payload is ignored (customers cannot set it). Body: {"phone": "+91 98765 43210", "company_name": "Postman Pvt Ltd", "tax_id": "HACK123"}	200	200	PASS
TC-056	Staff lists customers with search	GET	/api/v1/customers?search=pm.{run_id}	Staff (INTERNAL)	Search matches name or e-mail.	200	200	PASS
TC-057	Customer cannot list customers	GET	/api/v1/customers	Customer 1	Staff-only list.	403	403	PASS
TC-058	Staff reads a customer by id	GET	/api/v1/customers/{cust1_id}	Staff (INTERNAL)	Staff may open any customer.	200	200	PASS
TC-059	Customer reads own record by id	GET	/api/v1/customers/{cust1_id}	Customer 1	Own record is allowed.	200	200	PASS
TC-060	Customer cannot read another customer	GET	/api/v1/customers/{cust2_id}	Customer 1	Returns 404 (not 403) so ids cannot be probed.	404	404	PASS
TC-061	Staff updates customer incl. tax id	PUT	/api/v1/customers/{cust1_id}	Staff (INTERNAL)	Staff can set tax_id. Body: {"tax_id": "29ABCDE1234F1Z5"}	200	200	PASS
TC-062	Customer cannot use staff update	PUT	/api/v1/customers/{cust1_id}	Customer 1	Staff-only endpoint. Body: {"tax_id": "X"}	403	403	PASS
TC-063	Unknown customer	GET	/api/v1/customers/999999	Staff (INTERNAL)	Unknown id.	404	404	PASS

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-064	Search with SQL-injection text is harmless	GET	/api/v1/customers?search=' OR 1=1 --	Staff (INTERNAL)	Queries are parameterised: no error and no data leak.	200	200	PASS

04 Taxes

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-065	Admin creates tax GST 18%	POST	/api/v1/taxes	Admin	Tax created. Body: {"name": "GST-{{run_id}}", "rate": "18.00"}	201	201	PASS
TC-066	Create tax - duplicate name (case-insensitive)	POST	/api/v1/taxes	Admin	Tax names are unique. Body: {"name": "gst-{{run_id}}", "rate": "5.00"}	409	409	PASS
TC-067	Create tax - rate above 100 rejected	POST	/api/v1/taxes	Admin	Rate must be 0-100. Body: {"name": "Bad-{{run_id}}", "rate": "101"}	422	422	PASS
TC-068	Create tax - negative rate rejected	POST	/api/v1/taxes	Admin	Rate must be 0-100. Body: {"name": "Neg-{{run_id}}", "rate": "-1"}	422	422	PASS
TC-069	Create tax - missing name rejected	POST	/api/v1/taxes	Admin	Required field. Body: {"rate": "5"}	422	422	PASS
TC-070	Create tax - staff forbidden	POST	/api/v1/taxes	Staff (INTERNAL)	Only admin sets up taxes. Body: {"name": "S-{{run_id}}", "rate": "5"}	403	403	PASS
TC-071	Create tax - customer forbidden	POST	/api/v1/taxes	Customer 1	Only admin sets up taxes. Body: {"name": "C-{{run_id}}", "rate": "5"}	403	403	PASS
TC-072	Staff lists taxes	GET	/api/v1/taxes?is_active=true	Staff (INTERNAL)	Staff can read taxes.	200	200	PASS
TC-073	Customer cannot list taxes	GET	/api/v1/taxes	Customer 1	Staff-only.	403	403	PASS
TC-074	Get tax by id	GET	/api/v1/taxes/{tax_id}	Staff (INTERNAL)	Single tax.	200	200	PASS
TC-075	Get unknown tax	GET	/api/v1/taxes/999999	Staff (INTERNAL)	Unknown id.	404	404	PASS
TC-076	Create throw-away tax	POST	/api/v1/taxes	Admin	Used for update/delete tests. Body: {"name": "TMP-{{run_id}}", "rate": "5.00"}	201	201	PASS
TC-077	Update tax rate	PUT	/api/v1/taxes/{tmp_tax_id}	Admin	Rate changed. Body: {"rate": "5.50"}	200	200	PASS
TC-078	Delete (deactivate) tax	DELETE	/api/v1/taxes/{tmp_tax_id}	Admin	Soft delete: no content returned.	204	204	PASS
TC-079	Deactivated tax is flagged inactive	GET	/api/v1/taxes/{tmp_tax_id}	Staff (INTERNAL)	Record kept, inactive.	200	200	PASS

05 Products and Variants

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-080	Admin creates product A (ERP Suite, taxed)	POST	/api/v1/products	Admin	Product created with the GST tax linked. Body: {"name": "ERP Suite {{run_id}}", "product_type": "SERVICE", "sales_price": "1000.00", "cost_price": "400.00", "notes": "Annual ERP licence",	201	201	PASS

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-081	Admin creates product B (Support Add-on, untaxed)	POST	/api/v1/products	Admin	Second product used for add-on lines. Body: {"name": "Support Add-on {{run_id}}", "product_type": "DIGITAL", "sales_price": "250.00", "cost_price": "100.00"}	201	201	PASS
TC-082	Create product - negative price rejected	POST	/api/v1/products	Admin	Price must be >= 0. Body: {"name": "Neg", "sales_price": "-5"}	422	422	PASS
TC-083	Create product - invalid type rejected	POST	/api/v1/products	Admin	Type is an enumeration. Body: {"name": "X", "product_type": "HARDWARE", "sales_price": "5"}	422	422	PASS
TC-084	Create product - empty name rejected	POST	/api/v1/products	Admin	Name is required. Body: {"name": "", "sales_price": "5"}	422	422	PASS
TC-085	Create product - unknown tax id rejected	POST	/api/v1/products	Admin	Tax must exist and be active. Body: {"name": "X", "sales_price": "5", "tax_ids": [999999]}	400	400	PASS
TC-086	Create product - staff forbidden	POST	/api/v1/products	Staff (INTERNAL)	Only admin manages the catalogue. Body: {"name": "X", "sales_price": "5"}	403	403	PASS
TC-087	Create product - customer forbidden	POST	/api/v1/products	Customer 1	Only admin manages the catalogue. Body: {"name": "X", "sales_price": "5"}	403	403	PASS
TC-088	Create product - no token	POST	/api/v1/products	Anonymous (no token)	Authentication required. Body: {"name": "X", "sales_price": "5"}	401	401	PASS
TC-089	Add variant Brand=Odoo (+560)	POST	/api/v1/products/{product_a_id}/variants	Admin	Final price = sales price + 560 (the example from the problem statement). Body: {"variant_name": "Odoo", "additional_price": "560.00", "options": [{"option_name": "Brand", "option_value": "Odoo"}]}	201	201	PASS
TC-090	Add variant - duplicate name rejected	POST	/api/v1/products/{product_a_id}/variants	Admin	Variant names are unique per product (case-insensitive). Body: {"variant_name": "odoo", "additional_price": "1.00"}	409	409	PASS
TC-091	Add variant - negative extra price rejected	POST	/api/v1/products/{product_a_id}/variants	Admin	Extra price must be >= 0. Body: {"variant_name": "Neg", "additional_price": "-1"}	422	422	PASS
TC-092	Add variant to unknown product	POST	/api/v1/products/999999/variants	Admin	Unknown product. Body: {"variant_name": "X", "additional_price": "1"}	404	404	PASS
TC-093	Add second variant Brand=SAP (+900)	POST	/api/v1/products/{product_a_id}/variants	Admin	A product can offer several variants. Body: {"variant_name": "SAP", "additional_price": "900.00", "options": [{"option_name": "Brand", "option_value": "SAP"}]}	201	201	PASS
TC-094	Update variant price	PUT	/api/v1/products/{product_a_id}/variants/{variant_sap_id}	Admin	Extra price changed; final price recalculated. Body: {"additional_price": "950.00"}	200	200	PASS
TC-095	Delete (deactivate) unused variant	DELETE	/api/v1/products/{product_a_id}/variants/{variant_sap_id}	Admin	Soft delete.	204	204	PASS
TC-096	Staff sees product with cost price and inactive variant	GET	/api/v1/products?search=ERP Suite {run_id}	Staff (INTERNAL)	Staff view includes cost price and inactive variants.	200	200	PASS
TC-097	Customer sees product without cost price or inactive variants	GET	/api/v1/products?search=ERP Suite {run_id}	Customer 1	Customer view hides internal data.	200	200	PASS
TC-098	Customer cannot see inactive products	GET	/api/v1/products?is_active=false&search=ERP Suite {run_id}	Customer 1	Active filter does not override customer visibility.	200	200	PASS
TC-099	Get product by id	GET	/api/v1/products/{product_a_id}	Staff (INTERNAL)	Single product with variants.	200	200	PASS

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-100	Update product notes	PUT	/api/v1/products/{product_a_id}	Admin	Partial update. Body: {"notes": "Annual ERP licence - updated"}	200	200	PASS
TC-101	Get unknown product	GET	/api/v1/products/999999	Staff (INTERNAL)	Unknown id.	404	404	PASS
TC-102	Create throw-away product	POST	/api/v1/products	Admin	For delete test. Body: {"name": "Temp {{run_id}}", "sales_price": "1.00"}	201	201	PASS
TC-103	Delete unused product	DELETE	/api/v1/products/{tmp_product_id}	Admin	Soft delete.	204	204	PASS
TC-104	Deleted product no longer found	GET	/api/v1/products/{tmp_product_id}	Staff (INTERNAL)	Hidden after deletion.	404	404	PASS

06 Plans

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-105	Plan 1 - Monthly Pro (pausable, renewable, ends in 1 year)	POST	/api/v1/plans	Admin	Plan with a product line; Odoo variant price flows into the effective unit price. Body: {"name": "Monthly Pro {{run_id}}", "billing_interval": "MONTHLY", "interval_count": 1, "minimum_quantity": 1, "end_date": "{{d_plus_365}}",	201	201	PASS
TC-106	Plan 2 - Basic Fixed (no pause, no renewal, no early close, no end date)	POST	/api/v1/plans	Admin	Restrictive plan for negative lifecycle tests. Body: {"name": "Basic Fixed {{run_id}}", "billing_interval": "MONTHLY", "allow_pause": false, "allow_renewal": false, "allow_early_close": false}	201	201	PASS
TC-107	Plan 3 - Flexible (early close allowed, price override 200)	POST	/api/v1/plans	Admin	Plan whose line overrides the product price. Body: {"name": "Flexible {{run_id}}", "billing_interval": "MONTHLY", "end_date": "{{d_plus_365}}", "allow_early_close": true, "allow_renewal": tru	201	201	PASS
TC-108	Plan 4 - Future plan (available from +30 days)	POST	/api/v1/plans	Admin	Availability window in the future. Body: {"name": "Future {{run_id}}", "billing_interval": "YEARLY", "start_date": "{{d_plus_30}}", "end_date": "{{d_plus_365}}"}	201	201	PASS
TC-109	Create plan - end date before start date	POST	/api/v1/plans	Admin	Date range validation. Body: {"name": "Bad dates", "start_date": "{{d_plus_30}}", "end_date": "{{d_minus_5}}"}	422	422	PASS
TC-110	Create plan - invalid interval	POST	/api/v1/plans	Admin	Only DAILY/WEEKLY/MONTHLY/YEARLY. Body: {"name": "Hourly", "billing_interval": "HOURLY"}	422	422	PASS
TC-111	Create plan - interval_count 0 rejected	POST	/api/v1/plans	Admin	Must be >= 1. Body: {"name": "Zero", "interval_count": 0}	422	422	PASS
TC-112	Create plan - minimum quantity 0 rejected	POST	/api/v1/plans	Admin	Must be >= 1. Body: {"name": "ZeroQty", "minimum_quantity": 0}	422	422	PASS
TC-113	Create plan - unknown product rejected	POST	/api/v1/plans	Admin	Plan lines must reference real products. Body: {"name": "Ghost", "products": [{"product_id": 999999}]}	400	400	PASS
TC-114	Create plan - duplicate product line rejected	POST	/api/v1/plans	Admin	Same product/variant twice. Body: {"name": "Dup lines", "products": [{"product_id": "__B__"}, {"product_id": "__B__"}]}	400	400	PASS
TC-115	Create plan - staff forbidden	POST	/api/v1/plans	Staff (INTERNAL)	Only admin creates plans. Body: {"name": "X"}	403	403	PASS

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-116	Create plan - customer forbidden	POST	/api/v1/plans	Customer 1	Only admin creates plans. Body: {"name": "X"}	403	403	PASS
TC-117	Staff sees future plan	GET	/api/v1/plans?search=Future {run_id}	Staff (INTERNAL)	Staff list includes plans outside their availability window.	200	200	PASS
TC-118	Customer does not see future plan in list	GET	/api/v1/plans?search=Future {run_id}	Customer 1	Customers only see plans available today.	200	200	PASS
TC-119	Customer cannot open future plan	GET	/api/v1/plans/{plan4_id}	Customer 1	Not yet available to customers.	404	404	PASS
TC-120	Customer sees available plan	GET	/api/v1/plans/{plan1_id}	Customer 1	Available plan is visible.	200	200	PASS
TC-121	Staff gets plan details	GET	/api/v1/plans/{plan1_id}	Staff (INTERNAL)	Plan with product lines.	200	200	PASS
TC-122	Update plan - end before start rejected	PUT	/api/v1/plans/{plan1_id}	Admin	Validated against the stored dates. Body: {"start_date": "{{d_plus_365}}", "end_date": "{{d_plus_30}}"}	400	400	PASS
TC-123	Update plan name	PUT	/api/v1/plans/{plan1_id}	Admin	Partial update. Body: {"minimum_quantity": 1, "name": "Monthly Pro {{run_id}}"}	200	200	PASS
TC-124	Get unknown plan	GET	/api/v1/plans/999999	Staff (INTERNAL)	Unknown id.	404	404	PASS
TC-125	Create throw-away plan	POST	/api/v1/plans	Admin	For delete test. Body: {"name": "Temp {{run_id}}"}	201	201	PASS
TC-126	Delete (deactivate) plan	DELETE	/api/v1/plans/{tmp_plan_id}	Admin	Soft delete.	204	204	PASS
TC-127	Deactivated plan is flagged inactive	GET	/api/v1/plans/{tmp_plan_id}	Staff (INTERNAL)	Kept for history.	200	200	PASS
TC-128	Delete product used by an active plan	DELETE	/api/v1/products/{product_b_id}	Admin	Protected while an active plan uses it.	409	409	PASS
TC-129	Delete variant used by an active plan	DELETE	/api/v1/products/{product_a_id}/variants/{variant_odoo_id}	Admin	Protected while an active plan uses it.	409	409	PASS

07 Quotation Templates

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-130	Admin creates template (15 days)	POST	/api/v1/quotation-templates	Admin	Template with plan and a product line. Body: {"name": "ERP Starter {{run_id}}", "plan_id": "__PLAN1__", "validity_days": 15, "description": "2 ERP seats with Odoo", "lines": [{"product_	201	201	PASS
TC-131	Create template - validity 0 rejected	POST	/api/v1/quotation-templates	Admin	Validity 1-365 days. Body: {"name": "Bad", "validity_days": 0}	422	422	PASS
TC-132	Create template - validity 366 rejected	POST	/api/v1/quotation-templates	Admin	Validity 1-365 days. Body: {"name": "Bad", "validity_days": 366}	422	422	PASS
TC-133	Create template - unknown plan rejected	POST	/api/v1/quotation-templates	Admin	Plan must exist. Body: {"name": "Bad", "plan_id": 999999}	400	400	PASS
TC-134	Create template - unknown product rejected	POST	/api/v1/quotation-templates	Admin	Product must exist. Body: {"name": "Bad", "lines": [{"product_id": 999999}]}	400	400	PASS
TC-135	Create template - staff forbidden	POST	/api/v1/quotation-templates	Staff (INTERNAL)	Admin-only. Body: {"name": "X"}	403	403	PASS
TC-136	Create template - customer forbidden	POST	/api/v1/quotation-templates	Customer 1	Admin-only. Body: {"name": "X"}	403	403	PASS
TC-137	Staff lists templates	GET	/api/v1/quotation-templates	Staff (INTERNAL)	Sales reps can browse templates.	200	200	PASS
TC-138	Customer cannot list templates	GET	/api/v1/quotation-templates	Customer 1	Internal feature.	403	403	PASS

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-139	Staff gets template	GET	/api/v1/quotation-templates/{tpl_id}	Staff (INTERNAL)	Single template.	200	200	PASS
TC-140	Get unknown template	GET	/api/v1/quotation-templates/999999	Staff (INTERNAL)	Unknown id.	404	404	PASS
TC-141	Update template description	PUT	/api/v1/quotation-templates/{tpl_id}	Admin	Partial update. Body: {"description": "2 ERP seats with Odoo - updated"}	200	200	PASS
TC-142	Create throw-away template	POST	/api/v1/quotation-templates	Admin	For delete test. Body: {"name": "Temp {{run_id}}"}	201	201	PASS
TC-143	Delete (deactivate) template	DELETE	/api/v1/quotation-templates/{tmp_tpl_id}	Admin	Soft delete.	204	204	PASS
TC-144	Deactivated template flagged inactive	GET	/api/v1/quotation-templates/{tmp_tpl_id}	Staff (INTERNAL)	Kept.	200	200	PASS
TC-145	Quotation from inactive template rejected	POST	/api/v1/quotations	Staff (INTERNAL)	Inactive templates cannot be used. Body: {"customer_id": "__C1__", "template_id": "__TMPTPL__"}	400	400	PASS

08 Quotations

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-146	Staff creates quotation from template (no lines supplied)	POST	/api/v1/quotations	Staff (INTERNAL)	Lines are copied from the template; Odoo variant => 1560 x 2; GST 18% added. Body: {"customer_id": "__C1__", "template_id": "__TPL__", "notes": "Internal: priority customer"}	201	201	PASS
TC-147	Create quotation with manual lines and line discount	POST	/api/v1/quotations	Staff (INTERNAL)	(1560 x 2 - 100 discount) = 3020; GST 18% = 543.60; total 3563.60. Body: {"customer_id": "__C1__", "plan_id": "__PLAN1__", "valid_until": "{{d_plus_30}}", "lines": [{"product_id": "__A__", "variant_id": "__ODOO__"	201	201	PASS
TC-148	Update draft quotation lines (reprices)	PUT	/api/v1/quotations/{q2_id}	Staff (INTERNAL)	Quantity 1 and no discount: 1560 + 280.80 tax = 1840.80. Body: {"lines": [{"product_id": "__A__", "variant_id": "__ODOO__", "quantity": 1}]}	200	200	PASS
TC-149	Customer cannot see a DRAFT quotation	GET	/api/v1/quotations/{q2_id}	Customer 1	Drafts are internal.	404	404	PASS
TC-150	Customer list never contains drafts	GET	/api/v1/quotations?page_size=100	Customer 1	Draft quotations are filtered out for customers.	200	200	PASS
TC-151	Delete draft quotation	DELETE	/api/v1/quotations/{q2_id}	Staff (INTERNAL)	Drafts can be deleted.	204	204	PASS
TC-152	Deleted quotation is gone	GET	/api/v1/quotations/{q2_id}	Staff (INTERNAL)	Soft-deleted.	404	404	PASS
TC-153	Create quotation - no lines and no template	POST	/api/v1/quotations	Staff (INTERNAL)	At least one line needed. Body: {"customer_id": "__C1__", "plan_id": "__PLAN1__"}	400	400	PASS
TC-154	Create quotation - unknown customer	POST	/api/v1/quotations	Staff (INTERNAL)	Customer must exist. Body: {"customer_id": 999999, "plan_id": "__PLAN1__", "lines": [{"product_id": "__B__"}]}	404	404	PASS
TC-155	Create quotation - valid_until in the past	POST	/api/v1/quotations	Staff (INTERNAL)	Cannot be dated in the past. Body: {"customer_id": "__C1__", "plan_id": "__PLAN1__", "valid_until": "{{d_minus_5}}", "lines": [{"product_id": "__B__"}]}	400	400	PASS

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-156	Create quotation - line discount larger than line	POST	/api/v1/quotations	Staff (INTERNAL)	Discount cannot exceed amount. Body: {"customer_id": "__C1__", "plan_id": "__PLAN1__", "lines": [{"product_id": "__B__", "discount_amount": "9999.00"}]}	400	400	PASS
TC-157	Create quotation - quantity 0	POST	/api/v1/quotations	Staff (INTERNAL)	Quantity >= 1. Body: {"customer_id": "__C1__", "plan_id": "__PLAN1__", "lines": [{"product_id": "__B__", "quantity": 0}]}	422	422	PASS
TC-158	Create quotation - unknown product	POST	/api/v1/quotations	Staff (INTERNAL)	Product must exist. Body: {"customer_id": "__C1__", "plan_id": "__PLAN1__", "lines": [{"product_id": 999999}]}	400	400	PASS
TC-159	Create quotation - variant of another product	POST	/api/v1/quotations	Staff (INTERNAL)	Variant must belong to product. Body: {"customer_id": "__C1__", "plan_id": "__PLAN1__", "lines": [{"product_id": "__B__", "variant_id": "__ODOO__"}]}	400	400	PASS
TC-160	Create quotation - customer forbidden	POST	/api/v1/quotations	Customer 1	Staff-only. Body: {"customer_id": "__C1__", "plan_id": "__PLAN1__", "lines": [{"product_id": "__B__"}]}	403	403	PASS
TC-161	Create quotation - no token	POST	/api/v1/quotations	Anonymous (no token)	Authentication required. Body: {"customer_id": "__C1__", "plan_id": "__PLAN1__", "lines": [{"product_id": "__B__"}]}	401	401	PASS
TC-162	Send quotation	POST	/api/v1/quotations/{q1_id}/send	Staff (INTERNAL)	DRAFT -> SENT.	200	200	PASS
TC-163	Send quotation again rejected	POST	/api/v1/quotations/{q1_id}/send	Staff (INTERNAL)	Only drafts can be sent.	400	400	PASS
TC-164	Edit a sent quotation rejected	PUT	/api/v1/quotations/{q1_id}	Staff (INTERNAL)	Only drafts can be edited. Body: {"notes": "too late"}	400	400	PASS
TC-165	Delete a sent quotation rejected	DELETE	/api/v1/quotations/{q1_id}	Staff (INTERNAL)	Only drafts can be deleted.	400	400	PASS
TC-166	Customer sees own sent quotation without internal notes	GET	/api/v1/quotations/{q1_id}	Customer 1	Internal notes are hidden from customers.	200	200	PASS
TC-167	Another customer cannot see it	GET	/api/v1/quotations/{q1_id}	Customer 2	Isolation between customers.	404	404	PASS
TC-168	Another customer cannot confirm it	POST	/api/v1/quotations/{q1_id}/confirm	Customer 2	Isolation between customers.	404	404	PASS
TC-169	Customer confirms quotation	POST	/api/v1/quotations/{q1_id}/confirm	Customer 1	SENT -> CONFIRMED.	200	200	PASS
TC-170	Confirm twice rejected	POST	/api/v1/quotations/{q1_id}/confirm	Customer 1	Only sent quotations can be confirmed.	400	400	PASS
TC-171	Create quotation for rejection	POST	/api/v1/quotations	Staff (INTERNAL)	Will be rejected by the customer. Body: {"customer_id": "__C1__", "plan_id": "__PLAN2__", "lines": [{"product_id": "__B__", "quantity": 1}]}	201	201	PASS
TC-172	Send quotation for rejection	POST	/api/v1/quotations/{q_rej_id}/send	Staff (INTERNAL)	DRAFT -> SENT.	200	200	PASS
TC-173	Customer rejects quotation	POST	/api/v1/quotations/{q_rej_id}/reject	Customer 1	SENT -> REJECTED.	200	200	PASS
TC-174	Reject twice rejected	POST	/api/v1/quotations/{q_rej_id}/reject	Customer 1	Only sent quotations can be rejected.	400	400	PASS
TC-175	Confirm after rejection rejected	POST	/api/v1/quotations/{q_rej_id}/confirm	Customer 1	Terminal state.	400	400	PASS
TC-176	Quotation 3 for customer 1 (Plan 2, 4 x Support)	POST	/api/v1/quotations	Staff (INTERNAL)	4 x 250, no tax = 1000.00. Body: {"customer_id": "__C1__", "plan_id": "__PLAN2__", "lines": [{"product_id": "__B__", "quantity": 4}]}	201	201	PASS
TC-177	Send quotation 3	POST	/api/v1/quotations/{q3_id}/send	Staff (INTERNAL)	DRAFT -> SENT.	200	200	PASS
TC-178	Customer confirms quotation 3	POST	/api/v1/quotations/{q3_id}/confirm	Customer 1	SENT -> CONFIRMED.	200	200	PASS

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-179	Quotation 4 for customer 2 (Plan 3, price override)	POST	/api/v1/quotations	Staff (INTERNAL)	No unit_price supplied: the plan's override (200) is used, 2 x 200 = 400. Body: {"customer_id": "__C2__", "plan_id": "__PLAN3__", "lines": [{"product_id": "__B__", "quantity": 2}]}	201	201	PASS
TC-180	Send quotation 4	POST	/api/v1/quotations/{q4_id}/send	Staff (INTERNAL)	DRAFT -> SENT.	200	200	PASS
TC-181	Customer 2 confirms quotation 4	POST	/api/v1/quotations/{q4_id}/confirm	Customer 2	SENT -> CONFIRMED.	200	200	PASS
TC-182	Quotation 5 for customer 1 (to be cancelled later)	POST	/api/v1/quotations	Staff (INTERNAL)	Third subscription source. Body: {"customer_id": "__C1__", "plan_id": "__PLAN2__", "lines": [{"product_id": "__B__", "quantity": 1}]}	201	201	PASS
TC-183	Send quotation 5	POST	/api/v1/quotations/{q5_id}/send	Staff (INTERNAL)	DRAFT -> SENT.	200	200	PASS
TC-184	Customer confirms quotation 5	POST	/api/v1/quotations/{q5_id}/confirm	Customer 1	SENT -> CONFIRMED.	200	200	PASS
TC-185	Staff filters quotations by status	GET	/api/v1/quotations?status=CONFIRMED&page_size=100	Staff (INTERNAL)	Status filter.	200	200	PASS
TC-186	Staff searches quotation by number	GET	/api/v1/quotations?search={q1_number}	Staff (INTERNAL)	Search on quotation number.	200	200	PASS
TC-187	Staff filters quotations by customer	GET	/api/v1/quotations?customer_id={cust2_id}	Staff (INTERNAL)	Customer filter.	200	200	PASS
TC-188	Get unknown quotation	GET	/api/v1/quotations/999999	Staff (INTERNAL)	Unknown id.	404	404	PASS

09 Subscriptions

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-189	Create subscription from a rejected quotation	POST	/api/v1/subscriptions	Staff (INTERNAL)	Only confirmed quotations can become subscriptions. Body: {"quotation_id": "__Q_REJ__"}	400	400	PASS
TC-190	Create subscription from unknown quotation	POST	/api/v1/subscriptions	Staff (INTERNAL)	Unknown quotation. Body: {"quotation_id": 999999}	404	404	PASS
TC-191	Create subscription - payment terms above 365 days	POST	/api/v1/subscriptions	Staff (INTERNAL)	Validation of payment terms. Body: {"quotation_id": "__Q1__", "payment_terms_days": 400}	422	422	PASS
TC-192	Create subscription 1 from quotation 1 (customer 1, Plan 1)	POST	/api/v1/subscriptions	Staff (INTERNAL)	Status CONFIRMED, lines copied from the quotation, number SUB-YYYY-NNNNNN. Body: {"quotation_id": "__Q1__", "payment_terms_days": 15}	201	201	PASS
TC-193	Duplicate subscription for the same quotation	POST	/api/v1/subscriptions	Staff (INTERNAL)	One live subscription per quotation. Body: {"quotation_id": "__Q1__"}	409	409	PASS
TC-194	Customer cannot create subscriptions	POST	/api/v1/subscriptions	Customer 1	Staff-only. Body: {"quotation_id": "__Q1__"}	403	403	PASS
TC-195	Create subscription 2 (customer 1, Plan 2, 4 x Support)	POST	/api/v1/subscriptions	Staff (INTERNAL)	Restrictive plan: no pause, no renewal, no early close. Body: {"quotation_id": "__Q3__", "payment_terms_days": 30}	201	201	PASS
TC-196	Create subscription 3 (customer 2, Plan 3)	POST	/api/v1/subscriptions	Staff (INTERNAL)	Customer 2's subscription, payment terms default to 0 days. Body: {"quotation_id": "__Q4__"}	201	201	PASS
TC-197	Create subscription 4 (customer 1, Plan 2) for cancellation test	POST	/api/v1/subscriptions	Staff (INTERNAL)	Will be cancelled while still CONFIRMED. Body: {"quotation_id": "__Q5__"}	201	201	PASS
TC-198	Customer cannot activate	POST	/api/v1/subscriptions/{sub1_id}/activate	Customer 1	Staff-only.	403	403	PASS

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-199	Activate with a start date in the past	POST	/api/v1/subscriptions/{sub2_id}/activate	Staff (INTERNAL)	Start date cannot be in the past. Body: {"start_date": "{{d_minus_5}}"}	400	400	PASS
TC-200	Activate subscription 1	POST	/api/v1/subscriptions/{sub1_id}/activate	Staff (INTERNAL)	CONFIRMED -> ACTIVE; expiry = plan end date; next billing date = start date.	200	200	PASS
TC-201	Activate an already active subscription	POST	/api/v1/subscriptions/{sub1_id}/activate	Staff (INTERNAL)	Only CONFIRMED subscriptions can be activated.	400	400	PASS
TC-202	Activate subscription 2 (Plan 2, no end date)	POST	/api/v1/subscriptions/{sub2_id}/activate	Staff (INTERNAL)	No expiry date, so it can never be renewed.	200	200	PASS
TC-203	Activate subscription 3 (customer 2)	POST	/api/v1/subscriptions/{sub3_id}/activate	Staff (INTERNAL)	ACTIVE with plan end date.	200	200	PASS
TC-204	Customer 1 reads own subscription	GET	/api/v1/subscriptions/{sub1_id}	Customer 1	Owner may read.	200	200	PASS
TC-205	Customer 2 cannot read customer 1's subscription	GET	/api/v1/subscriptions/{sub1_id}	Customer 2	Isolation between customers (404 not 403).	404	404	PASS
TC-206	Staff reads any subscription	GET	/api/v1/subscriptions/{sub1_id}	Staff (INTERNAL)	Staff access.	200	200	PASS
TC-207	Customer list contains only own subscriptions	GET	/api/v1/subscriptions?page_size=100	Customer 1	Row-level isolation.	200	200	PASS
TC-208	Staff filters by status ACTIVE	GET	/api/v1/subscriptions?status=ACTIVE&page_size=100	Staff (INTERNAL)	Status filter.	200	200	PASS
TC-209	Staff searches by subscription number	GET	/api/v1/subscriptions?search={sub1_number}	Staff (INTERNAL)	Search on number.	200	200	PASS
TC-210	Staff filters by customer	GET	/api/v1/subscriptions?customer_id={cust2_id}	Staff (INTERNAL)	Customer filter.	200	200	PASS
TC-211	Unknown subscription	GET	/api/v1/subscriptions/999999	Staff (INTERNAL)	Unknown id.	404	404	PASS
TC-212	Customer pauses subscription 1	POST	/api/v1/subscriptions/{sub1_id}/pause	Customer 1	Plan allows pausing: ACTIVE -> PAUSED.	200	200	PASS
TC-213	Invoice cannot be generated while paused	POST	/api/v1/invoices/generate	Staff (INTERNAL)	Only ACTIVE subscriptions are invoiced. Body: {"subscription_id": "__SUB1__"}	400	400	PASS
TC-214	Customer resumes subscription 1	POST	/api/v1/subscriptions/{sub1_id}/resume	Customer 1	PAUSED -> ACTIVE.	200	200	PASS
TC-215	Resume a subscription that is not paused	POST	/api/v1/subscriptions/{sub1_id}/resume	Customer 1	Only PAUSED subscriptions can be resumed.	400	400	PASS
TC-216	Pause blocked by plan rule	POST	/api/v1/subscriptions/{sub2_id}/pause	Customer 1	Plan 2 does not allow pausing.	400	400	PASS
TC-217	Customer 2 cannot pause someone else's subscription	POST	/api/v1/subscriptions/{sub1_id}/pause	Customer 2	Isolation between customers.	404	404	PASS
TC-218	Close blocked: plan does not allow early close	POST	/api/v1/subscriptions/{sub1_id}/close	Staff (INTERNAL)	Expiry date not reached and plan has allow_early_close = false.	400	400	PASS
TC-219	Customer requests cancellation	POST	/api/v1/subscriptions/{sub1_id}/request-cancel	Customer 1	Request recorded for staff to action. Body: {"reason": "Too expensive"}	200	200	PASS
TC-220	Duplicate cancellation request	POST	/api/v1/subscriptions/{sub1_id}/request-cancel	Customer 1	Only one pending request at a time.	409	409	PASS
TC-221	Staff finds subscriptions with pending requests	GET	/api/v1/subscriptions?has_request=true&page_size=100	Staff (INTERNAL)	Work-queue filter.	200	200	PASS
TC-222	Staff dismisses the request	POST	/api/v1/subscriptions/{sub1_id}/dismiss-request	Staff (INTERNAL)	Request declined, subscription unchanged. Body: {"reason": "Customer retained"}	200	200	PASS
TC-223	Dismiss when nothing is pending	POST	/api/v1/subscriptions/{sub1_id}/dismiss-request	Staff (INTERNAL)	No pending request.	400	400	PASS
TC-224	Staff cannot use the customer request endpoint	POST	/api/v1/subscriptions/{sub1_id}/request-cancel	Staff (INTERNAL)	Customer-only endpoint.	403	403	PASS

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-225	Customer 2 cannot request on another's subscription	POST	/api/v1/subscriptions/{sub1_id}/request-cancel	Customer 2	Isolation between customers.	404	404	PASS
TC-226	Customer requests renewal	POST	/api/v1/subscriptions/{sub1_id}/request-renew	Customer 1	Renewal request pending.	200	200	PASS
TC-227	Renewal request blocked by plan rule	POST	/api/v1/subscriptions/{sub2_id}/request-renew	Customer 1	Plan 2 does not allow renewal.	400	400	PASS
TC-228	Staff renews subscription 1 for 2 periods	POST	/api/v1/subscriptions/{sub1_id}/renew	Staff (INTERNAL)	Expiry extended by 2 months; pending request cleared. Body: {"periods": 2}	200	200	PASS
TC-229	Renew blocked: plan does not allow renewal	POST	/api/v1/subscriptions/{sub2_id}/renew	Staff (INTERNAL)	Plan 2 disallows renewal.	400	400	PASS
TC-230	Customer cannot renew directly	POST	/api/v1/subscriptions/{sub1_id}/renew	Customer 1	Staff-only.	403	403	PASS
TC-231	Staff reads status history	GET	/api/v1/subscriptions/{sub1_id}/history	Staff (INTERNAL)	Every change is audited with who did it.	200	200	PASS
TC-232	Customer sees history without internal user ids	GET	/api/v1/subscriptions/{sub1_id}/history	Customer 1	changed_by is hidden from customers.	200	200	PASS
TC-233	Customer cannot cancel directly	POST	/api/v1/subscriptions/{sub4_id}/cancel	Customer 1	Staff-only.	403	403	PASS
TC-234	Staff cancels a CONFIRMED subscription	POST	/api/v1/subscriptions/{sub4_id}/cancel	Staff (INTERNAL)	CONFIRMED -> CANCELLED, billing stopped. Body: {"reason": "Customer changed mind"}	200	200	PASS
TC-235	Cancel an already cancelled subscription	POST	/api/v1/subscriptions/{sub4_id}/cancel	Staff (INTERNAL)	Terminal state.	400	400	PASS

10 Discounts

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-236	Create percentage discount 10%	POST	/api/v1/discounts	Admin	Code is normalised to upper case; starts unused, applies to every product. Body: {"code": "ten{{run_id}}", "name": "Ten percent", "discount_type": "PERCENTAGE", "discount_value": "10"}	201	201	PASS
TC-237	Create fixed discount 50 (usage limit 1)	POST	/api/v1/discounts	Admin	Single-use code. Body: {"code": "FIXONCE{{run_id}}", "name": "Flat 50 once", "discount_type": "FIXED", "discount_value": "50.00", "usage_limit": 1}	201	201	PASS
TC-238	Create discount with minimum purchase 100000	POST	/api/v1/discounts	Admin	For the minimum-purchase check. Body: {"code": "MINBIG{{run_id}}", "name": "Big spenders", "discount_type": "PERCENTAGE", "discount_value": "20", "min_purchase_amount": "100000.0	201	201	PASS
TC-239	Create already-expired discount	POST	/api/v1/discounts	Admin	Date window entirely in the past. Body: {"code": "OLD{{run_id}}", "name": "Expired", "discount_type": "PERCENTAGE", "discount_value": "15", "start_date": "{{d_minus_10}}", "end_dat	201	201	PASS
TC-240	Create product-specific discount (Support Add-on only)	POST	/api/v1/discounts	Admin	Mapped to one product. Body: {"code": "PRODB{{run_id}}", "name": "Support only", "discount_type": "PERCENTAGE", "discount_value": "10", "product_ids": ["__PRODB__"]}	201	201	PASS
TC-241	Create throw-away discount	POST	/api/v1/discounts	Admin	For deactivation test. Body: {"code": "TMPD{{run_id}}", "name": "Temp", "discount_type": "FIXED", "discount_value": "5.00"}	201	201	PASS

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-242	Create discount - duplicate code	POST	/api/v1/discounts	Admin	Codes are unique. Body: {"code": "TEN{{run_id}}", "name": "Dup", "discount_type": "PERCENTAGE", "discount_value": "5"}	409	409	PASS
TC-243	Create discount - percentage above 100	POST	/api/v1/discounts	Admin	Percent <= 100. Body: {"code": "BIG{{run_id}}", "name": "Big", "discount_type": "PERCENTAGE", "discount_value": "150"}	422	422	PASS
TC-244	Create discount - zero value	POST	/api/v1/discounts	Admin	Value must be > 0. Body: {"code": "ZERO{{run_id}}", "name": "Zero", "discount_type": "FIXED", "discount_value": "0"}	422	422	PASS
TC-245	Create discount - invalid code characters	POST	/api/v1/discounts	Admin	Letters, digits, '-' or '_' only. Body: {"code": "BAD CODE!", "name": "Bad", "discount_type": "FIXED", "discount_value": "5"}	422	422	PASS
TC-246	Create discount - end before start	POST	/api/v1/discounts	Admin	Date range validation. Body: {"code": "DATES{{run_id}}", "name": "Dates", "discount_type": "FIXED", "discount_value": "5", "start_date": "{{d_plus_30}}", "end_date": "{{	422	422	PASS
TC-247	Create discount - usage limit 0	POST	/api/v1/discounts	Admin	Usage limit >= 1. Body: {"code": "LIM{{run_id}}", "name": "Lim", "discount_type": "FIXED", "discount_value": "5", "usage_limit": 0}	422	422	PASS
TC-248	Create discount - unknown product	POST	/api/v1/discounts	Admin	Mapped products must exist. Body: {"code": "GHOST{{run_id}}", "name": "Ghost", "discount_type": "FIXED", "discount_value": "5", "product_ids": [999999]}	400	400	PASS
TC-249	Create discount - staff forbidden	POST	/api/v1/discounts	Staff (INTERNAL)	Only the admin can create discount rules. Body: {"code": "STAFF{{run_id}}", "name": "S", "discount_type": "FIXED", "discount_value": "5"}	403	403	PASS
TC-250	Create discount - customer forbidden	POST	/api/v1/discounts	Customer 1	Only the admin can create discount rules. Body: {"code": "CUST{{run_id}}", "name": "C", "discount_type": "FIXED", "discount_value": "5"}	403	403	PASS
TC-251	Staff lists discounts	GET	/api/v1/discounts?search={run_id}	Staff (INTERNAL)	Staff can read discount rules.	200	200	PASS
TC-252	Customer cannot list discounts	GET	/api/v1/discounts	Customer 1	Staff-only.	403	403	PASS
TC-253	Staff gets a discount	GET	/api/v1/discounts/{ten_id}	Staff (INTERNAL)	Single rule.	200	200	PASS
TC-254	Get unknown discount	GET	/api/v1/discounts/999999	Staff (INTERNAL)	Unknown id.	404	404	PASS
TC-255	Admin updates discount name	PUT	/api/v1/discounts/{ten_id}	Admin	Partial update. Body: {"name": "Ten percent (updated)"}	200	200	PASS
TC-256	Update discount - percentage above 100	PUT	/api/v1/discounts/{ten_id}	Admin	Re-validated on update. Body: {"discount_value": "250"}	400	400	PASS
TC-257	Staff cannot update discount	PUT	/api/v1/discounts/{ten_id}	Staff (INTERNAL)	Admin-only. Body: {"name": "x"}	403	403	PASS
TC-258	Admin deactivates throw-away discount	DELETE	/api/v1/discounts/{tmpd_id}	Admin	Soft delete.	204	204	PASS
TC-259	Deactivated discount is flagged inactive	GET	/api/v1/discounts/{tmpd_id}	Staff (INTERNAL)	Record kept.	200	200	PASS

11 Billing and Invoices

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-260	Generate invoice for subscription 1	POST	/api/v1/invoices/generate	Staff (INTERNAL)	Invoice built from the subscription: lines, GST 18%, totals, payment terms 15 days. Body: {"subscription_id": "__SUB1__"}	201	201	PASS
TC-261	Second manual generation for the same period is refused	POST	/api/v1/invoices/generate	Staff (INTERNAL)	Next billing date moved forward; nothing is due again. Body: {"subscription_id": "__SUB1__"}	400	400	PASS
TC-262	Invoice for a cancelled subscription is refused	POST	/api/v1/invoices/generate	Staff (INTERNAL)	Only ACTIVE subscriptions can be invoiced. Body: {"subscription_id": "__SUB4__"}	400	400	PASS
TC-263	Invoice for unknown subscription	POST	/api/v1/invoices/generate	Staff (INTERNAL)	Unknown id. Body: {"subscription_id": 999999}	404	404	PASS
TC-264	Customer cannot generate invoices	POST	/api/v1/invoices/generate	Customer 1	Staff-only. Body: {"subscription_id": "__SUB1__"}	403	403	PASS
TC-265	Billing run (billing day)	POST	/api/v1/billing/run	Staff (INTERNAL)	All due ACTIVE subscriptions get a draft invoice in one job (subscriptions 2 and 3 are due today).	200	200	PASS
TC-266	Billing created exactly one invoice for subscription 2	GET	/api/v1/invoices?subscription_id={sub2_id}	Staff (INTERNAL)	Invoice built from 4 x 250, untaxed.	200	200	PASS
TC-267	Billing created exactly one invoice for subscription 3	GET	/api/v1/invoices?subscription_id={sub3_id}	Staff (INTERNAL)	Invoice built from 2 x 200 (plan price override).	200	200	PASS
TC-268	Billing run again - nothing is billed twice	POST	/api/v1/billing/run	Staff (INTERNAL)	Idempotent: no new invoices for already-billed periods.	200	200	PASS
TC-269	Subscription 2 still has exactly one invoice	GET	/api/v1/invoices?subscription_id={sub2_id}	Staff (INTERNAL)	Unique billing period per subscription.	200	200	PASS
TC-270	Billing run with a future date	POST	/api/v1/billing/run	Staff (INTERNAL)	run_date cannot be in the future. Body: {"run_date": "{{d_plus_5}}"}	400	400	PASS
TC-271	Billing run with an earlier date creates nothing new	POST	/api/v1/billing/run	Staff (INTERNAL)	Nothing was due on that earlier date. Body: {"run_date": "{{d_minus_5}}"}	200	200	PASS
TC-272	List billing jobs	GET	/api/v1/billing/jobs	Staff (INTERNAL)	Audit trail of every run (newest first).	200	200	PASS
TC-273	Customer cannot run billing	POST	/api/v1/billing/run	Customer 1	Staff-only.	403	403	PASS
TC-274	Customer cannot list billing jobs	GET	/api/v1/billing/jobs	Customer 1	Staff-only.	403	403	PASS
TC-275	Close subscription 3 early (plan allows it)	POST	/api/v1/subscriptions/{sub3_id}/close	Staff (INTERNAL)	ACTIVE -> CLOSED on a plan with allow_early_close = true.	200	200	PASS
TC-276	Closed subscription cannot be invoiced	POST	/api/v1/invoices/generate	Staff (INTERNAL)	Only ACTIVE subscriptions are invoiced. Body: {"subscription_id": "__SUB3__"}	400	400	PASS
TC-277	Customer 2 cannot see a DRAFT invoice	GET	/api/v1/invoices/{inv_c_id}	Customer 2	Drafts are internal.	404	404	PASS
TC-278	Customer list never contains drafts	GET	/api/v1/invoices?page_size=100	Customer 2	Draft invoices are filtered out.	200	200	PASS
TC-279	Customer 1 cannot see draft invoice A	GET	/api/v1/invoices/{inv_a_id}	Customer 1	Drafts are internal.	404	404	PASS
TC-280	Staff edits draft invoice (notes and due date)	PUT	/api/v1/invoices/{inv_a_id}	Staff (INTERNAL)	Draft invoices can be adjusted before confirming. Body: {"notes": "Net 30 agreed", "due_date": "{{d_plus_30}}"}	200	200	PASS
TC-281	Edit invoice - due date before invoice date	PUT	/api/v1/invoices/{inv_a_id}	Staff (INTERNAL)	Date sanity check. Body: {"due_date": "{{d_minus_10}}"}	400	400	PASS
TC-282	Customer cannot edit invoices	PUT	/api/v1/invoices/{inv_a_id}	Customer 1	Staff-only. Body: {"notes": "x"}	403	403	PASS
TC-283	Invoice history starts with DRAFT	GET	/api/v1/invoices/{inv_a_id}/history	Staff (INTERNAL)	Audit trail.	200	200	PASS
TC-284	Customer cannot apply a discount	POST	/api/v1/invoices/{inv_a_id}/apply-discount	Customer 1	Staff-only. Body: {"code": "{{ten_code}}"}	403	403	PASS

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-285	Apply unknown discount code	POST	/api/v1/invoices/{inv_a_id}/apply-discount	Staff (INTERNAL)	Code must exist. Body: {"code": "NOSUCHCODE"}	400	400	PASS
TC-286	Apply deactivated discount code	POST	/api/v1/invoices/{inv_a_id}/apply-discount	Staff (INTERNAL)	Inactive codes stop working. Body: {"code": "{{tmpd_code}}"}	400	400	PASS
TC-287	Apply expired discount code	POST	/api/v1/invoices/{inv_a_id}/apply-discount	Staff (INTERNAL)	Discounts stop when they expire. Body: {"code": "{{old_code}}"}	400	400	PASS
TC-288	Apply discount below minimum purchase	POST	/api/v1/invoices/{inv_a_id}/apply-discount	Staff (INTERNAL)	Minimum purchase not met. Body: {"code": "{{minbig_code}}"}	400	400	PASS
TC-289	Apply product-specific discount to an invoice without that product	POST	/api/v1/invoices/{inv_a_id}/apply-discount	Staff (INTERNAL)	Invoice A only contains product A. Body: {"code": "{{prodb_code}}"}	400	400	PASS
TC-290	Apply 10% discount to invoice A	POST	/api/v1/invoices/{inv_a_id}/apply-discount	Staff (INTERNAL)	10% of 3120 = 312.00; tax recalculated on 2808 = 505.44; total 3313.44. Body: {"code": "{{ten_code}}"}	200	200	PASS
TC-291	Second discount on the same invoice refused	POST	/api/v1/invoices/{inv_a_id}/apply-discount	Staff (INTERNAL)	One discount per invoice. Body: {"code": "{{fix_code}}"}	409	409	PASS
TC-292	Remove discount from invoice A	POST	/api/v1/invoices/{inv_a_id}/remove-discount	Staff (INTERNAL)	Totals revert to the undiscounted values.	200	200	PASS
TC-293	Remove discount when none applied	POST	/api/v1/invoices/{inv_a_id}/remove-discount	Staff (INTERNAL)	Nothing to remove.	400	400	PASS
TC-294	Re-apply 10% discount to invoice A	POST	/api/v1/invoices/{inv_a_id}/apply-discount	Staff (INTERNAL)	Final invoice A total: 3313.44. Body: {"code": "{{ten_code}}"}	200	200	PASS
TC-295	Discount usage counter increased	GET	/api/v1/discounts/{ten_id}	Staff (INTERNAL)	used_count tracks real usage.	200	200	PASS
TC-296	Apply product-specific discount to invoice B	POST	/api/v1/invoices/{inv_b_id}/apply-discount	Staff (INTERNAL)	10% of 1000 = 100.00 on the Support line. Body: {"code": "{{prodb_code}}"}	200	200	PASS
TC-297	Remove it from invoice B	POST	/api/v1/invoices/{inv_b_id}/remove-discount	Staff (INTERNAL)	Back to 1000.00.	200	200	PASS
TC-298	Apply single-use fixed discount to invoice B	POST	/api/v1/invoices/{inv_b_id}/apply-discount	Staff (INTERNAL)	Flat 50 off: 950.00. Uses up the code (limit 1). Body: {"code": "{{fix_code}}"}	200	200	PASS
TC-299	Same code refused once its usage limit is reached	POST	/api/v1/invoices/{inv_c_id}/apply-discount	Staff (INTERNAL)	Discounts stop when they run out. Body: {"code": "{{fix_code}}"}	400	400	PASS
TC-300	Remove the single-use discount from invoice B	POST	/api/v1/invoices/{inv_b_id}/remove-discount	Staff (INTERNAL)	Releases the use again.	200	200	PASS
TC-301	Released discount can be used again (counter back to 0)	GET	/api/v1/discounts/{fix_id}	Staff (INTERNAL)	used_count decremented.	200	200	PASS
TC-302	Pay a draft invoice is refused	POST	/api/v1/payments	Staff (INTERNAL)	Payments only on confirmed/overdue/part-paid invoices. Body: {"method": "CASH", "invoice_id": "__INVC__", "amount": "10.00"}	400	400	PASS
TC-303	Confirm invoice A	POST	/api/v1/invoices/{inv_a_id}/confirm	Staff (INTERNAL)	DRAFT -> CONFIRMED; customer is notified.	200	200	PASS
TC-304	Confirm invoice A twice	POST	/api/v1/invoices/{inv_a_id}/confirm	Staff (INTERNAL)	Only drafts can be confirmed.	400	400	PASS
TC-305	Edit a confirmed invoice	PUT	/api/v1/invoices/{inv_a_id}	Staff (INTERNAL)	Only drafts can be edited. Body: {"notes": "late"}	400	400	PASS
TC-306	Apply discount to a confirmed invoice	POST	/api/v1/invoices/{inv_a_id}/apply-discount	Staff (INTERNAL)	Only drafts can be discounted. Body: {"code": "{{fix_code}}"}	400	400	PASS
TC-307	Remove discount from a confirmed invoice	POST	/api/v1/invoices/{inv_a_id}/remove-discount	Staff (INTERNAL)	Only drafts can be changed.	400	400	PASS
TC-308	Confirm invoice B	POST	/api/v1/invoices/{inv_b_id}/confirm	Staff (INTERNAL)	DRAFT -> CONFIRMED.	200	200	PASS
TC-309	Confirm invoice C	POST	/api/v1/invoices/{inv_c_id}/confirm	Staff (INTERNAL)	DRAFT -> CONFIRMED.	200	200	PASS

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-310	Customer 1 reads confirmed invoice A (no internal notes)	GET	/api/v1/invoices/{inv_a_id}	Customer 1	Customers see their confirmed invoices but never internal notes.	200	200	PASS
TC-311	Customer 2 cannot read customer 1's invoice	GET	/api/v1/invoices/{inv_a_id}	Customer 2	Isolation between customers.	404	404	PASS
TC-312	Customer 1 cannot read customer 2's invoice	GET	/api/v1/invoices/{inv_c_id}	Customer 1	Isolation between customers.	404	404	PASS
TC-313	Customer invoice list is scoped to the customer	GET	/api/v1/invoices?page_size=100	Customer 1	Only own, non-draft invoices.	200	200	PASS
TC-314	Invoice PDF (staff)	GET	/api/v1/invoices/{inv_a_id}/pdf	Staff (INTERNAL)	Downloadable PDF document.	200	200	PASS
TC-315	Invoice PDF (owning customer)	GET	/api/v1/invoices/{inv_a_id}/pdf	Customer 1	Customer can download own invoice.	200	200	PASS
TC-316	Invoice PDF (other customer)	GET	/api/v1/invoices/{inv_a_id}/pdf	Customer 2	Isolation between customers.	404	404	PASS
TC-317	Invoice PDF (no token)	GET	/api/v1/invoices/{inv_a_id}/pdf	Anonymous (no token)	Authentication required.	401	401	PASS
TC-318	Staff filters invoices by status	GET	/api/v1/invoices?status=CONFIRMED&page_size=100	Staff (INTERNAL)	Status filter.	200	200	PASS
TC-319	Staff searches invoice by number	GET	/api/v1/invoices?search={inv_a_number}	Staff (INTERNAL)	Search on invoice number.	200	200	PASS
TC-320	Staff filters invoices by customer	GET	/api/v1/invoices?customer_id={cust2_id}&page_size=100	Staff (INTERNAL)	Customer filter.	200	200	PASS
TC-321	Unknown invoice	GET	/api/v1/invoices/999999	Staff (INTERNAL)	Unknown id.	404	404	PASS

12 Payments

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-322	Customer cannot record a payment	POST	/api/v1/payments	Customer 1	Staff-only. Body: {"method": "CASH", "invoice_id": "__INVA__", "amount": "10.00"}	403	403	PASS
TC-323	Payment without token	POST	/api/v1/payments	Anonymous (no token)	Authentication required. Body: {"method": "CASH", "invoice_id": "__INVA__", "amount": "10.00"}	401	401	PASS
TC-324	Payment with both invoice_id and allocations	POST	/api/v1/payments	Staff (INTERNAL)	Exactly one target allowed. Body: {"method": "CASH", "amount": "10.00", "invoice_id": "__INVA__", "allocations": [{"invoice_id": "__INVA__", "amount": "10.00"}]}	422	422	PASS
TC-325	Payment with neither invoice_id nor allocations	POST	/api/v1/payments	Staff (INTERNAL)	One target required. Body: {"method": "CASH", "amount": "10.00"}	422	422	PASS
TC-326	Payment with zero amount	POST	/api/v1/payments	Staff (INTERNAL)	Amount must be positive. Body: {"method": "CASH", "invoice_id": "__INVA__", "amount": "0"}	422	422	PASS
TC-327	Payment with invalid method	POST	/api/v1/payments	Staff (INTERNAL)	Unknown method. Body: {"method": "BITCOIN", "invoice_id": "__INVA__", "amount": "10.00"}	422	422	PASS
TC-328	Payment on unknown invoice	POST	/api/v1/payments	Staff (INTERNAL)	Unknown invoice. Body: {"method": "CASH", "invoice_id": 999999, "amount": "10.00"}	404	404	PASS

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-329	Overpayment is rejected	POST	/api/v1/payments	Staff (INTERNAL)	Amount above balance due is refused. Body: {"method": "CASH", "invoice_id": "__INVA__", "amount": "99999.00"}	400	400	PASS
TC-330	Split allocation not adding up to amount	POST	/api/v1/payments	Staff (INTERNAL)	Allocations must sum to amount. Body: {"method": "CASH", "amount": "100.00", "allocations": [{"invoice_id": "__INVA__", "amount": "40.00"}]}	400	400	PASS
TC-331	Split payment across invoices of different customers	POST	/api/v1/payments	Staff (INTERNAL)	One payment belongs to one customer. Body: {"method": "CASH", "amount": "20.00", "allocations": [{"invoice_id": "__INVA__", "amount": "10.00"}, {"invoice_id": "__INVC__", "amount": "1	400	400	PASS
TC-332	Same invoice twice in one payment	POST	/api/v1/payments	Staff (INTERNAL)	Duplicates refused. Body: {"method": "CASH", "amount": "20.00", "allocations": [{"invoice_id": "__INVA__", "amount": "10.00"}, {"invoice_id": "__INVA__", "amount": "1	400	400	PASS
TC-333	Wrong customer_id for the invoice	POST	/api/v1/payments	Staff (INTERNAL)	Customer must own the invoice. Body: {"method": "CASH", "invoice_id": "__INVA__", "amount": "10.00", "customer_id": "__C2__"}	400	400	PASS
TC-334	Partial payment of invoice A (1000.00, bank transfer)	POST	/api/v1/payments	Staff (INTERNAL)	Invoice A becomes PARTIALLY_PAID. Body: {"method": "BANK_TRANSFER", "invoice_id": "__INVA__", "amount": "1000.00", "payment_reference": "PM-{{run_id}}-P1", "note": "internal: first	201	201	PASS
TC-335	Invoice A is PARTIALLY_PAID with correct balance	GET	/api/v1/invoices/{inv_a_id}	Staff (INTERNAL)	paid 1000, balance 2313.44.	200	200	PASS
TC-336	Duplicate payment reference is refused	POST	/api/v1/payments	Staff (INTERNAL)	References are unique. Body: {"method": "CASH", "invoice_id": "__INVA__", "amount": "10.00", "payment_reference": "PM-{{run_id}}-P1"}	409	409	PASS
TC-337	Auto-generated reference when none is supplied (split payment A + B)	POST	/api/v1/payments	Staff (INTERNAL)	Split allocation across two invoices of customer 1; reference PAY-YYYY-NNNNNN. Body: {"method": "UPI", "amount": "1500.00", "allocations": [{"invoice_id": "__INVA__", "amount": "500.00"}, {"invoice_id": "__INVB__", "amount":	201	201	PASS
TC-338	Invoice B is now PAID	GET	/api/v1/invoices/{inv_b_id}	Staff (INTERNAL)	Fully settled by the split payment.	200	200	PASS
TC-339	Payment on a PAID invoice is refused	POST	/api/v1/payments	Staff (INTERNAL)	Settled invoices accept nothing. Body: {"method": "CASH", "invoice_id": "__INVB__", "amount": "1.00"}	400	400	PASS
TC-340	Pay the remainder of invoice A (1813.44)	POST	/api/v1/payments	Staff (INTERNAL)	Invoice A becomes PAID. Body: {"method": "CARD", "invoice_id": "__INVA__", "amount": "1813.44"}	201	201	PASS
TC-341	Invoice A is PAID	GET	/api/v1/invoices/{inv_a_id}	Staff (INTERNAL)	paid == total.	200	200	PASS
TC-342	Partial payment of invoice C (customer 2)	POST	/api/v1/payments	Staff (INTERNAL)	Invoice C PARTIALLY_PAID. Body: {"method": "CASH", "invoice_id": "__INVC__", "amount": "150.00"}	201	201	PASS
TC-343	Get payment (staff sees note)	GET	/api/v1/payments/{pay1_id}	Staff (INTERNAL)	Staff view includes the note.	200	200	PASS
TC-344	Get payment (owning customer: note hidden)	GET	/api/v1/payments/{pay1_id}	Customer 1	Customers never see internal notes.	200	200	PASS
TC-345	Get payment (other customer)	GET	/api/v1/payments/{pay1_id}	Customer 2	Isolation between customers.	404	404	PASS

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-346	Unknown payment	GET	/api/v1/payments/999999	Staff (INTERNAL)	Unknown id.	404	404	PASS
TC-347	Customer payment list is scoped	GET	/api/v1/payments?page_size=100	Customer 1	Only own payments.	200	200	PASS
TC-348	Staff filters payments by invoice	GET	/api/v1/payments?invoice_id={inv_a_id}	Staff (INTERNAL)	Invoice filter.	200	200	PASS
TC-349	Staff filters payments by method	GET	/api/v1/payments?method=UPI&page_size=100	Staff (INTERNAL)	Method filter.	200	200	PASS
TC-350	Staff searches payment by reference	GET	/api/v1/payments?search=PM-{run_id}-P1	Staff (INTERNAL)	Reference search.	200	200	PASS
TC-351	Customer cannot refund	POST	/api/v1/payments/{pay1_id}/refund	Customer 1	Staff-only.	403	403	PASS
TC-352	Refund payment 3 (card, final instalment)	POST	/api/v1/payments/{pay3_id}/refund	Staff (INTERNAL)	Allocations reversed, payment REFUNDED, reason stored. Body: {"reason": "Customer dispute"}	200	200	PASS
TC-353	Invoice A back to PARTIALLY_PAID after refund	GET	/api/v1/invoices/{inv_a_id}	Staff (INTERNAL)	paid 1500, balance 1813.44.	200	200	PASS
TC-354	Refund twice is refused	POST	/api/v1/payments/{pay3_id}/refund	Staff (INTERNAL)	Only COMPLETED payments.	400	400	PASS
TC-355	Pay again after refund (re-collect 1813.44)	POST	/api/v1/payments	Staff (INTERNAL)	Invoice A PAID again. Body: {"method": "CARD", "invoice_id": "__INVA__", "amount": "1813.44"}	201	201	PASS
TC-356	Payments list filtered to REFUNDED	GET	/api/v1/payments?status=REFUNDED&page_size=100	Staff (INTERNAL)	Status filter.	200	200	PASS

13 Notifications

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-357	Notifications without token	GET	/api/v1/notifications	Anonymous (no token)	Authentication required.	401	401	PASS
TC-358	Customer 1 has notifications (invoice / payment events)	GET	/api/v1/notifications?page_size=100	Customer 1	Customer received payment/invoice notifications.	200	200	PASS
TC-359	Unread count	GET	/api/v1/notifications/unread-count	Customer 1	Counter returned.	200	200	PASS
TC-360	Unread-only filter	GET	/api/v1/notifications?unread_only=true&page_size=100	Customer 1	Filter works.	200	200	PASS
TC-361	Mark one notification read	POST	/api/v1/notifications/{notif_id}/read	Customer 1	Status READ with read_at.	200	200	PASS
TC-362	Another customer cannot read my notification	POST	/api/v1/notifications/{notif_id}/read	Customer 2	Notifications are private.	404	404	PASS
TC-363	Unknown notification	POST	/api/v1/notifications/999999/read	Customer 1	Unknown id.	404	404	PASS
TC-364	Mark all read	POST	/api/v1/notifications/read-all	Customer 1	Bulk update.	200	200	PASS
TC-365	Unread count is zero afterwards	GET	/api/v1/notifications/unread-count	Customer 1	Zero.	200	200	PASS
TC-366	Customer cannot trigger overdue reminders	POST	/api/v1/notifications/send-overdue-reminders	Customer 1	Staff-only.	403	403	PASS
TC-367	Staff triggers overdue reminders	POST	/api/v1/notifications/send-overdue-reminders	Staff (INTERNAL)	Returns created count.	200	200	PASS

14 Reports and Dashboard

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-368	Summary (staff)	GET	/api/v1/reports/summary	Staff (INTERNAL)	All dashboard KPIs present.	200	200	PASS
TC-369	Summary (admin)	GET	/api/v1/reports/summary	Admin	Admin allowed too.	200	200	PASS
TC-370	Summary (customer)	GET	/api/v1/reports/summary	Customer 1	Reports are staff-only.	403	403	PASS
TC-371	Summary (no token)	GET	/api/v1/reports/summary	Anonymous (no token)	Authentication required.	401	401	PASS
TC-372	Revenue by month	GET	/api/v1/reports/revenue?group_by=month	Staff (INTERNAL)	Monthly points.	200	200	PASS
TC-373	Revenue by day with range	GET	/api/v1/reports/revenue?group_by=day&from={d_minus_10}&to={d_plus_5}	Staff (INTERNAL)	Daily points inside the window.	200	200	PASS
TC-374	Revenue invalid group_by	GET	/api/v1/reports/revenue?group_by=year	Staff (INTERNAL)	Only day or month.	422	422	PASS
TC-375	Revenue (customer)	GET	/api/v1/reports/revenue	Customer 1	Staff-only.	403	403	PASS
TC-376	Overdue invoices report	GET	/api/v1/reports/overdue	Staff (INTERNAL)	Paginated overdue list.	200	200	PASS
TC-377	Overdue report (customer)	GET	/api/v1/reports/overdue	Customer 1	Staff-only.	403	403	PASS

15 Security and Access Control

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-378	Anonymous cannot list customers	GET	/api/v1/customers	Anonymous (no token)	Authentication required on every protected route.	401	401	PASS
TC-379	Anonymous cannot list subscriptions	GET	/api/v1/subscriptions	Anonymous (no token)	Authentication required on every protected route.	401	401	PASS
TC-380	Anonymous cannot list invoices	GET	/api/v1/invoices	Anonymous (no token)	Authentication required on every protected route.	401	401	PASS
TC-381	Anonymous cannot list payments	GET	/api/v1/payments	Anonymous (no token)	Authentication required on every protected route.	401	401	PASS
TC-382	Anonymous cannot list products	GET	/api/v1/products	Anonymous (no token)	Authentication required on every protected route.	401	401	PASS
TC-383	Anonymous cannot list plans	GET	/api/v1/plans	Anonymous (no token)	Authentication required on every protected route.	401	401	PASS
TC-384	Anonymous cannot list users	GET	/api/v1/users	Anonymous (no token)	Authentication required on every protected route.	401	401	PASS
TC-385	Anonymous cannot list reports	GET	/api/v1/reports/summary	Anonymous (no token)	Authentication required on every protected route.	401	401	PASS
TC-386	Anonymous cannot list notifications	GET	/api/v1/notifications	Anonymous (no token)	Authentication required on every protected route.	401	401	PASS
TC-387	Garbage bearer token	GET	/api/v1/auth/me	garbage_token	Invalid token rejected.	401	401	PASS
TC-388	Customer cannot list users	GET	/api/v1/users	Customer 1	Admin-only.	403	403	PASS
TC-389	Staff cannot list users	GET	/api/v1/users	Staff (INTERNAL)	Admin-only.	403	403	PASS
TC-390	Customer cannot list customers	GET	/api/v1/customers	Customer 1	Staff-only.	403	403	PASS

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-391	Customer cannot create product	POST	/api/v1/products	Customer 1	Admin-only write. Body: {"name": "Hack", "product_type": "SERVICE", "sales_price": "1.00"}	403	403	PASS
TC-392	Staff cannot create tax	POST	/api/v1/taxes	Staff (INTERNAL)	Admin-only write. Body: {"name": "Hack tax", "rate": "5.00"}	403	403	PASS
TC-393	Staff cannot create discount	POST	/api/v1/discounts	Staff (INTERNAL)	Admin-only write. Body: {"name": "Hack", "code": "HACK", "discount_type": "PERCENTAGE", "discount_value": "10"}	403	403	PASS
TC-394	Customer cannot run billing	POST	/api/v1/billing/run	Customer 1	Staff-only.	403	403	PASS
TC-395	Admin can read staff-level data	GET	/api/v1/invoices?page_size=5	Admin	Admin inherits staff access.	200	200	PASS
TC-396	SQL-injection style search is harmless	GET	/api/v1/customers?search=' OR 1=1 --	Staff (INTERNAL)	Parameterised queries: no error, no data leak.	200	200	PASS
TC-397	Oversized page_size is rejected or capped	GET	/api/v1/invoices?page_size=100000	Staff (INTERNAL)	Pagination bounds prevent unbounded queries.	422	422	PASS

16 E-mail Token Flows (manual tokens)

ID	Title	Method	Endpoint	Role	Expected result	Exp.	Act.	Result
TC-398	Verify e-mail with real token	POST	/api/v1/auth/verify-email	Anonymous (no token)	Marks the e-mail verified. Body: {"token": "{{verify_token}}"}	200	200	PASS*
TC-399	Verify e-mail token cannot be reused	POST	/api/v1/auth/verify-email	Anonymous (no token)	Single use. Body: {"token": "{{verify_token}}"}	400	400	PASS*
TC-400	Reset password with real token	POST	/api/v1/auth/reset-password	Anonymous (no token)	Password changed. Body: {"token": "{{reset_token}}", "new_password": "New#Strong9Pass"}	200	200	PASS*
TC-401	Reset token cannot be reused	POST	/api/v1/auth/reset-password	Anonymous (no token)	Single use. Body: {"token": "{{reset_token}}", "new_password": "Another#Strong9"}	400	400	PASS*